

Solve place-value problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A city has 3,804,215 people. Another has 398,760 fewer. What is the second population? Copy or complete the first step.

2. A lift moves from level -5 to level 18. How many floors does it travel?

3. What is 100,000 more than 6,945,120?

4. Miro says: Miro changes more than one digit when adding 100,000 to 6,945,120. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A city has 3,804,215 people. Another has 398,760 fewer. What is the second population? Copy or complete the first step.
Answer 3,405,455.
2. A lift moves from level -5 to level 18. How many floors does it travel?
Answer 23 floors.
3. What is 100,000 more than 6,945,120?
Answer 7,045,120.
4. Miro says: Miro changes more than one digit when adding 100,000 to 6,945,120. Is this correct? Explain.
Answer Add one hundred-thousand and regroup only when needed. The answer is 7,045,120.

Solve place-value problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A city has 3,804,215 people. Another has 398,760 fewer. What is the second population?

6. Check this result using an inverse, estimate or known fact: What is 100,000 more than 6,945,120?

2. A lift moves from level -5 to level 18. How many floors does it travel?

7. A number rounds to 4,000,000 to the nearest million. Give the smallest possible whole number.

3. What is 100,000 more than 6,945,120?

8. Identify and correct this misconception: Miro changes more than one digit when adding 100,000 to 6,945,120.

4. Solve this independently and show every stage: A city has 3,804,215 people. Another has 398,760 fewer. What is the second population?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A lift moves from level -5 to level 18. How many floors does it travel?

10. Write and solve a similar question to: What is 100,000 more than 6,945,120?

Teacher answers and checking prompts

1. A city has 3,804,215 people. Another has 398,760 fewer. What is the second population?
Answer 3,405,455.
2. A lift moves from level -5 to level 18. How many floors does it travel?
Answer 23 floors.
3. What is 100,000 more than 6,945,120?
Answer 7,045,120.
4. Solve this independently and show every stage: A city has 3,804,215 people. Another has 398,760 fewer. What is the second population?
Answer 3,405,455.
5. Use a second method or representation for: A lift moves from level -5 to level 18. How many floors does it travel?
Answer Expected result: 23 floors. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: What is 100,000 more than 6,945,120?
Answer Expected result: 7,045,120. A valid check must be shown.
7. A number rounds to 4,000,000 to the nearest million. Give the smallest possible whole number.
Answer 3,500,000.
8. Identify and correct this misconception: Miro changes more than one digit when adding 100,000 to 6,945,120.
Answer Add one hundred-thousand and regroup only when needed. The answer is 7,045,120.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: What is 100,000 more than 6,945,120?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve place-value problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. A number rounds to 4,000,000 to the nearest million. Give the smallest possible whole number.

6. Create a new example that tests the same idea as: What is 100,000 more than 6,945,120?

2. Prove the answer to this question, rather than only calculating it: A city has 3,804,215 people. Another has 398,760 fewer. What is the second population?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A lift moves from level -5 to level 18. How many floors does it travel?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 7,045,120.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes more than one digit when adding 100,000 to 6,945,120.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A number rounds to 4,000,000 to the nearest million. Give the smallest possible whole number.
Answer 3,500,000.
2. Prove the answer to this question, rather than only calculating it: A city has 3,804,215 people. Another has 398,760 fewer. What is the second population?
Answer Expected result: 3,405,455. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A lift moves from level -5 to level 18. How many floors does it travel?
Answer Expected result: 23 floors. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 7,045,120.
Answer One valid reconstruction is: What is 100,000 more than 6,945,120? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes more than one digit when adding 100,000 to 6,945,120.
Answer Add one hundred-thousand and regroup only when needed. The answer is 7,045,120.
6. Create a new example that tests the same idea as: What is 100,000 more than 6,945,120?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.